



Manual on “Data and information on sensor deployment”

This manual provides step-by-step instructions for submitting **data and information describing how and when the data was collected**.

Please, help us completing the form. The information you provide helps researchers, regulatory bodies, and decision-making systems better understand the data and ensure its quality, reliability, traceability, and comparability.

<https://landsealot.s4oceandata.eu/>

Complete this form at the end of each measurement carried out with an instrument (device).

A measurement is defined as a data collection activity that:

- is carried out using the same instrument (device),
- takes place at a specific point/area/transect in time,
- is performed by the same person or by the same group of people.

The measurement could be:

- Continuous measurement (set up the sensor and then leave),
- Occasional measurement (take the measurement while present).

1. Information about the user

First, indicate whether you are a **scientist** part of the LandSeaLot project or a **citizen**.

If you are a scientist, provide information about yourself and any collaborators involved.

If you are a **citizen**, you will be asked whether you agree to have your name (and of any collaborators) associated with the data you collected, or whether you prefer to remain anonymous. This information matters because the data you collect may be used in scientific publications, reports, or public databases and if you agree, your name will publicly appear, giving you formal recognition for your work. If choose to remain anonymous, your data will still be used, but your name will not be disclosed.

If you agree, you consent to the processing of your personal data (first name and surname) in accordance with Regulation (EU) 2016/679 (GDPR), solely for project-related purposes linked to the collection of environmental data and their sharing with the scientific community through open access repositories for research and publication purposes.

If you are a scientist or a citizen who agrees to be identified, please provide **Surname, Name**, and **ORCID** (if available) - a unique identifier used in the scientific community to unambiguously associate researchers with their work. If you have one, you can find it at <https://orcid.org>. If you do not have one, you may leave this field blank.

If other people joined the activity, please provide the same information for each **contributor**.



2. Description of the device

You must select the **sensor** you used (e.g., "EnvLogger").

If you do not know the name of the sensor you used, you can refer to the pictures in the form next to each sensor's name.

3. Description of the measurement

Upload the data file

Upload the data file following the instructions provided in the form specific to the sensor you used.

General description

You are asked to describe the **activity** carried out, including: **what** data was collected, **how** it was collected, **where** the measurement took place, and **who** was involved. For example: *"The sampling took place in Loco Beach as part of a citizen science activity. The sensor was mounted on a rock face to continuously collect data at that specific location for 10 hours..."*

Location and depth

First, indicate the **country** where the measurement took place. If you are in the UK, specify whether the activity took place inside a marina (if you do not know, please ask to the activity supervisor).

Then, specify the **spatial deployment** of the device:

- **Fixed location:** provide geographic coordinates (latitude and longitude) in WGS84 format. These can be obtained using a GPS device or Google Maps: click on the desired point on the map and copy the coordinates - they are already in the correct format (e.g. 44.12345, 8.98765).;
- **Area or along a trajectory (no GPS device available):** manually enter the working area by indicating the minimum and maximum latitude and longitude values in WGS84 format. These can be obtained using Google Maps as described above;
- **Area or along a trajectory (GPS device available):** upload the GPX file. The GPS device must record positional data in GPX format, which will later be used to associate each measurement with its corresponding geographic coordinates.

Ensure that the GPS device is configured in UTC time zone. To check or change this setting, go to the device's settings and look for a 'Time' or 'System' section. If your GPS does not allow time zone configuration, ensure it records time in UTC and note this for post-processing.

Start the measurement with GPS device and sensor at the same time.

Finally, indicate whether the measurements were taken at **one fixed depth** or at **different depths** and enter the depth value in meters.



Platform – ONLY FOR SCIENTISTS

A platform is any physical or technological system that supports and enables the collection of measurements in a given environment, such as vessels, fixed stations, structures, or sampling points.

Select the platform from the list. If the platform type you used is not in the list, select “Other” and indicate it according to NERC vocabulary L06 <https://vocab.nerc.ac.uk/collection/L06/current/>.

If applicable, you may also enter a **pre-existing platform identifier** (e.g. an official name or code assigned to the vessel or station) and any **additional information about the deployment** that may be useful for data interpretation.

If the platform is a boat/ship/vessel, please indicate its identifiers (IMO number and maritime call sign).

The list of possible platforms:

- beach/intertidal zone structure: A structure to which instrumentation may be attached that is either in air or under water depending on the state of the tide and weather conditions.
- coastal structure: A fixed man-made structure permanently linked to land with access to water at all states of the tide to which instrumentation may be attached (e.g. pier).
- fishing vessel: A platform operating on the surface of the water column whose primary purpose is the commercial harvesting of fish or shellfish but may be engaged in scientific activities such as fish stock surveys or mooring deployments and recoveries.
- fixed subsurface vertical profiler: A platform that periodically makes an automated vertical traverse of the water column at a predetermined fixed location. (e.g. YSI vertical profiler, HOMER CTD).
- float: A free-floating platform either on the surface of the water column or at a predetermined depth within the water column.
- land or seafloor: A platform located on the solid surface of the Earth either above or below sea level.
- land/onshore structure: A fixed man-made structure on land to which instrumentation may be attached (e.g. meteorological tower).
- man-powered boat: A platform operating on the surface of the water column that is manually propelled and may not be easily removed from the water (e.g. trireme).
- moored surface buoy: An unmanned instrumented platform operating on the surface of the water column loosely tethered to the seafloor to maintain a fixed position (e.g. ODAS buoy).
- mooring: A tethered collection of oceanographic instruments at a fixed location that may include seafloor, mid-water and surface components.
- naval vessel: A platform operating on the surface of the water column in unpredictable locations that is primarily equipped, manned and operated for military purposes. Includes surface warships of all sizes and logistic support vessels.
- offshore structure: A fixed (for the duration of the measurements) man-made structure away from the coast to which instrumentation may be attached (e.g. oil rig, gas rig or jack-up barge).



- research vessel: A platform of any size operating on the surface of the water column in unpredictable locations that is specifically equipped, manned and operated for scientific, usually oceanographic, research.
- river station: An instrumented structure in a river upstream of its tidal limit.
- self-propelled boat: A small self-propelled platform operating on the surface of the water column in unpredictable locations that is smaller than a ship, but too large to easily remove from the water.
- ship: A large platform operating on the surface of the water column. Objective definitions for guidelines: >50m length (EU), >100 foot length (USA), >300 GRT weight (SOLAS). Subjective definition: a ship is a vessel big enough to carry a boat.
- subsurface mooring: A collection of oceanographic instruments attached to wires suspended between anchors on the seabed and buoyant spheres in the water column.
- surface vessel: A mobile platform with propulsion operating on and restricted to the surface of a water body.
- vessel at fixed position: A platform of any size occupying a fixed location on the surface of the water column for prolonged periods collecting scientific (oceanographic and meteorological) data as a primary or secondary mission. Includes light vessels and weather ships.
- vessel of opportunity: A platform for purpose of commerce of any size operating on the surface of the water column in unpredictable locations that regularly collects scientific (oceanographic and meteorological) data (e.g. an instrumented cargo vessel).
- vessel of opportunity on fixed route: A platform repeatedly following a predictable fixed track on the surface of the water column that collects scientific (oceanographic and meteorological) data (e.g. an instrumented ferry).
- Other: any other platform not indicated in the list according to the NERC L06 Vocabulary, <https://vocab.nerc.ac.uk/collection/L06/current/>.

Some of “Other”:

- autonomous surface water vehicle: A self-propelled vehicle operating on the sea surface with no human occupants.
- autonomous underwater vehicle: A free-roving platform operating in the water column with propulsion but no human operator on board (e.g. Autosub, Gavia).
- diver: A human being with self-contained equipment or surface-connected suit enabling operation within the water column.
- drifting manned submersible: A platform operating in the water column attached to a mothership by an umbilical but with no means of propulsion that has at least one human operator on board (e.g. bathysphere).
- drifting subsurface float: An unmanned instrumented platform drifting freely in the water column at a depth governed by its density (e.g. Swallow float).
- drifting subsurface profiling float: An unmanned instrumented platform drifting freely in the water column that periodically makes vertical traverses through the water column (e.g. Argo float).
- drifting surface float: An unmanned instrumented platform operating on the surface of



the water column often attached to a drogue to track currents rather than winds (e.g. Argos buoy).

- fixed benthic node: A collection of oceanographic instruments mounted at a fixed position on the seabed (e.g. POL Monitoring Platform, seabed ADCP).
- human: A human being without specialised equipment operating on land or the surface of the water column (i.e., it is the person who measures the data and interprets it, e.g. Midi Secchi).
- ice island: A floating ice sheet detached from the coast.
- ice shelf: A floating ice sheet attached to the coast.
- Ice-tethered subsurface profiling float: A platform that periodically makes an automated vertical traverse of the water column that is fixed to a floating body of ice.
- kite: An aerofoil tethered to the ground held aloft by the wind.
- lowered unmanned submersible: An unmanned platform lowered and raised vertically by a cable from the mothership. Includes any type of profiling sensor mounting such as CTD frames, profiling radiometers and instrumented nets.
- man-powered small boat: A platform operating on the surface of the water column that is manually propelled and may be easily removed from the water (e.g. rowing boat, canoe).
- propelled manned submersible: A platform operating in the water column that has both self-contained propulsion and at least one human operator on board (e.g. submarine).
- propelled unmanned submersible: A platform operating in the water column attached to a mothership by an umbilical with limited propulsion and no human operator on board (e.g. ROV).
- self-propelled small boat: A small self-propelled platform operating on the surface of the water column that may be easily removed from the water (e.g. shore-based RIBs, ships' boats).
- sub-surface gliders: Platforms with buoyancy-based propulsion that are capable of operations at variable depths which are not constrained to be near the sea surface.
- submersible: A platform operating within a water body.
- surface gliders: Platforms operating at a single depth near the sea surface, using a combination of solar energy and wave motion as means of propulsion.